

Global Supply Chain Shipment Performance Analysis

Project Overview

This project focuses on analyzing global healthcare supply chain shipment data to evaluate shipment performance, delivery reliability, freight cost behavior, transportation efficiency, and vendor performance.

The objective of this project is to identify operational inefficiencies, shipment delays, high-cost transportation patterns, vendor-related delivery issues, and product-level shipment contribution within the supply chain network.

The project was developed using:

- Python for data preprocessing, feature engineering, and exploratory data analysis (EDA)
- SQL (MySQL) for structured business querying and operational analysis
- Power BI for interactive dashboard creation and business visualization

The analysis answers important business questions such as:

- Which countries receive the highest number of shipments?
- Which shipment modes are used most frequently?
- Which vendors contribute to higher shipment delays?
- How freight costs vary across shipment methods?
- Which healthcare products contribute the highest shipment value?
- What factors influence shipment delivery performance?

This project helps organizations improve:

- logistics planning
 - transportation cost optimization
 - vendor monitoring
 - shipment efficiency
 - delivery reliability
 - operational decision-making
-

Dataset Summary

The dataset used in this project contains:

- 10,325 rows
- 33 columns

representing global healthcare supply chain shipment records.

The dataset contains operational shipment information such as:

- shipment IDs
- vendor information
- freight cost
- shipment mode
- delivery dates
- product details
- quantity and pricing
- shipment destinations
- transportation details
- delivery performance information

The dataset contains both:

Numerical Variables

- Freight Cost (USD)
- Weight (Kilograms)
- Line Item Quantity
- Line Item Value
- Pack Price
- Unit Price
- Delay Days

Categorical Variables

- Country
- Shipment Mode
- Vendor
- Product Group
- Delivery Status
- Vendor INCO Term
- Molecule/Test Type

Date Columns

- Scheduled Delivery Date
- Delivered to Client Date

- Delivery Recorded Date
- PO Sent to Vendor Date

The dataset also contained:

- missing values
- inconsistent date formatting
- operational shipment records
- multiple transportation methods

making it suitable for performing real-world supply chain analytics.

```
df = pd.read_csv("SCMS_Delivery_History_Dataset.csv")
```

```
df.head()
```

ID	Project Code	PQ #	PO / SO #	ASN/DN #	Country	Managed By	Fulfill Via	Vendor INCO Term	Shipment Mode	...	Unit of Measure (Per Pack)	Line Item Quantity	Line Item Value	Pack Price	Unit Price	Manufacturing Site	First Line Designation	Weight (Kilograms)	Freight Cost (USD)	Line Item Insurance (USD)
1	100-CL-T01	Pre-PQ Process	SCMS-4	ASN-8	Côte d'Ivoire	PMO - US	Direct Drop	EXW	Air	...	30	19	551.0	29.00	0.97	Ranbaxy Fine Chemicals LTD	Yes	13	780.34	NaN
3	108-VN-T01	Pre-PQ Process	SCMS-13	ASN-85	Vietnam	PMO - US	Direct Drop	EXW	Air	...	240	1000	6200.0	6.20	0.03	Aurobindo Unit III, India	Yes	358	4521.5	NaN
4	100-CL-T01	Pre-PQ Process	SCMS-20	ASN-14	Côte d'Ivoire	PMO - US	Direct Drop	FCA	Air	...	100	500	40000.0	80.00	0.80	ABBVIE GmbH & Co.KG Wiesbaden	Yes	171	1653.78	NaN
15	108-VN-T01	Pre-PQ Process	SCMS-78	ASN-50	Vietnam	PMO - US	Direct Drop	EXW	Air	...	60	31920	127360.8	3.99	0.07	Ranbaxy, Paonta Shahib, India	Yes	1855	16007.06	NaN
16	108-VN-T01	Pre-PQ Process	SCMS-81	ASN-55	Vietnam	PMO - US	Direct Drop	EXW	Air	...	60	38000	121600.0	3.20	0.05	Aurobindo Unit III, India	Yes	7590	45450.08	NaN

Exploratory Data Analysis (EDA) using Python

Exploratory Data Analysis (EDA) was performed using Python to understand shipment patterns, identify operational issues, analyze delivery performance, and prepare the dataset for business analysis and dashboard creation.

Python libraries used:

- pandas
- numpy
- matplotlib
- seaborn

The EDA process focused on:

- shipment distribution

- vendor analysis
 - freight cost analysis
 - delivery delay analysis
 - transportation analysis
 - product contribution analysis
 - country-wise shipment trends
-

Data Loading and Inspection

The dataset was loaded using pandas and inspected using various functions to understand the structure and quality of the dataset.

Functions used:

```
import pandas as pd
```

```
pd.read_csv()  
df.head()  
df.info()  
df.describe()
```

Initial inspection helped identify:

- total number of records
- column names
- data types
- missing values
- numerical statistics
- operational shipment information

The dataset initially contained multiple object-type date columns and missing operational values.

```
df.info()
✓ 0.0s

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10324 entries, 0 to 10323
Data columns (total 33 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                    10324 non-null  int64
1   Project Code                         10324 non-null  object
2   PQ #                                10324 non-null  object
3   PO / SO #                           10324 non-null  object
4   ASN/DN #                            10324 non-null  object
5   Country                             10324 non-null  object
6   Managed By                          10324 non-null  object
7   Fulfill Via                         10324 non-null  object
8   Vendor INCO Term                   10324 non-null  object
9   Shipment Mode                      9964 non-null   object
10  PQ First Sent to Client Date        10324 non-null  object
11  PO Sent to Vendor Date              10324 non-null  object
12  Scheduled Delivery Date             10324 non-null  object
13  Delivered to Client Date            10324 non-null  object
14  Delivery Recorded Date              10324 non-null  object
15  Product Group                      10324 non-null  object
16  Sub Classification                  10324 non-null  object
17  Vendor                             10324 non-null  object
18  Item Description                    10324 non-null  object
19  Molecule/Test Type                10324 non-null  object
...
31  Freight Cost (USD)                 10324 non-null  object
32  Line Item Insurance (USD)          10037 non-null  float64
dtypes: float64(4), int64(3), object(26)
memory usage: 2.6+ MB
```

Data Cleaning

Data cleaning was performed to improve dataset consistency, accuracy, and usability for analysis.

The following preprocessing steps were completed:

Data Type Conversion

The following date columns were converted into datetime format:

- Scheduled Delivery Date
- Delivered to Client Date
- Delivery Recorded Date
- PO Sent to Vendor Date
- PQ First Sent to Client Date

This helped perform shipment delay analysis and trend analysis.

Missing Value Analysis

Missing values were identified in columns such as:

- Shipment Mode
- Dosage
- Freight Cost (USD)
- PQ First Sent to Client Date
- PO Sent to Vendor Date

Some missing values were preserved because they represented unavailable operational data rather than incorrect records.

Duplicate Check

Duplicate records were checked using:

```
df.duplicated().sum()
```

No duplicate rows were found in the dataset.

Standardization

Shipment mode values and operational text fields were standardized to improve consistency.

```
df.isnull().sum()
```

✓ 0.0s

ID	0
Project Code	0
PQ #	0
PO / SO #	0
ASN/DN #	0
Country	0
Managed By	0
Fulfill Via	0
Vendor INCO Term	0
Shipment Mode	360
PQ First Sent to Client Date	205
PO Sent to Vendor Date	328
Scheduled Delivery Date	0
Delivered to Client Date	0
Delivery Recorded Date	0
Product Group	0
Sub Classification	0
Vendor	0
Item Description	0
Molecule/Test Type	0
Brand	0
Dosage	1736
Dosage Form	0
Unit of Measure (Per Pack)	0
Line Item Quantity	0
...	
First Line Designation	0
Weight (Kilograms)	1507
Freight Cost (USD)	1681
Line Item Insurance (USD)	287

dtype: int64

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings...](#)

```
date_cols = [  
    'PO Sent to Vendor Date',  
    'Scheduled Delivery Date',  
    'Delivered to Client Date',  
    'Delivery Recorded Date'  
]
```

✓ 0.0s

```
for col in date_cols:  
    df[col] = pd.to_datetime(df[col], errors='coerce')
```

✓ 0.6s

```
df.info()
✓ 0.0s

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10324 entries, 0 to 10323
Data columns (total 33 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                    10324 non-null  int64
1   Project Code                         10324 non-null  object
2   PQ #                                10324 non-null  object
3   PO / SO #                           10324 non-null  object
4   ASN/DN #                            10324 non-null  object
5   Country                             10324 non-null  object
6   Managed By                          10324 non-null  object
7   Fulfill Via                         10324 non-null  object
8   Vendor INCO Term                   10324 non-null  object
9   Shipment Mode                      9964 non-null   object
10  PQ First Sent to Client Date        7643 non-null   datetime64[ns]
11  PO Sent to Vendor Date              4592 non-null   datetime64[ns]
12  Scheduled Delivery Date            10324 non-null   datetime64[ns]
13  Delivered to Client Date           10324 non-null   datetime64[ns]
14  Delivery Recorded Date             10324 non-null   datetime64[ns]
15  Product Group                      10324 non-null   object
16  Sub Classification                 10324 non-null   object
17  Vendor                            10324 non-null   object
18  Item Description                   10324 non-null   object
19  Molecule/Test Type               10324 non-null   object
...
31  Freight Cost (USD)                 8643 non-null   object
32  Line Item Insurance (USD)          10037 non-null  float64
dtypes: datetime64[ns](5), float64(4), int64(3), object(21)
memory usage: 2.6+ MB

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Feature Engineering

New columns were created to support shipment performance analysis and dashboard development.

1. Delay Days

Delay Days was calculated using:

```
df['Delay Days'] = (
    df['Delivered to Client Date'] -
```



```
df['Scheduled Delivery Date']  
)dt.days
```

This feature measures the number of days a shipment was delayed or delivered early.

- Positive values → delayed shipments
- Negative values → early deliveries
- Zero → on-time delivery

2. Delivery Status

Delivery status categories were created based on Delay Days:

- Early
- On Time
- Late

This helped analyze shipment reliability.

3. Freight Cost Per KG

A new metric was created to evaluate transportation efficiency:

```
df['Freight Cost Per KG'] = \  
    df['Freight Cost (USD)'] / df['Weight (Kilograms)']
```

This feature helps compare freight efficiency across shipments.

```
df['Freight Cost Per KG'] = (  
    df['Freight Cost (USD)']  
    / df['Weight (Kilograms)']  
)  
✓ 0.0s
```

```
df['Delay Days'] = (  
    df['Delivered to Client Date']  
    - df['Scheduled Delivery Date']  
) .dt.days
```

✓ 0.0s

```
df['Delivery Status'] = df['Delay Days'].apply(  
    lambda x:  
        'Late' if x > 0  
        else ('Early' if x < 0 else 'On Time')  
)
```

✓ 0.0s

Data Understanding

Statistical analysis was performed to understand the distribution and behavior of important numerical variables.

Important variables analyzed:

- Freight Cost (USD)
- Weight (Kilograms)
- Line Item Value
- Delay Days
- Shipment Quantity

The analysis revealed:

- high variation in freight costs
- shipment delivery inconsistencies
- operational dependency on specific vendors
- uneven shipment distribution across countries

Summary statistics helped identify:

- minimum values
- maximum values
- average shipment cost
- shipment spread
- operational outliers

df.describe()

✓ 0.1s

C:\Users\HP\AppData\Roaming\Python\Python313\site-packages\pandas\core\nanops.py:1016: RuntimeWarning: invalid value encountered in subtract
sqr = _ensure_numeric((avg - values) ** 2)

	ID	PQ First Sent to Client Date	PO Sent to Vendor Date	Scheduled Delivery Date	Delivered to Client Date	Delivery Recorded Date	Unit of Measure (Per Pack)	Line Item Quantity	Line Item Value	Pack Price	Unit Price
count	10324.000000	7643	4592	10324	10324	10324	10324.000000	10324.000000	1.032400e+04	10324.000000	10324.000000
mean	51098.968229	2012-06-07 08:05:31.597540096	2012-01-25 07:21:13.170731776	2011-10-04 15:30:28.593568512	2011-09-28 14:56:34.963192576	2011-10-01 12:33:28.523827968	77.990895	18332.534870	1.576506e+05	21.910241	0.611701
min	1.000000	2009-01-07 00:00:00	2006-04-19 00:00:00	2006-05-02 00:00:00	2006-05-02 00:00:00	2006-05-02 00:00:00	1.000000	1.000000	0.000000e+00	0.000000	0.000000
25%	12795.750000	2010-11-26 00:00:00	2010-02-04 00:00:00	2009-09-08 00:00:00	2009-09-01 00:00:00	2009-09-01 00:00:00	30.000000	408.000000	4.314593e+03	4.120000	0.080000
50%	57540.500000	2012-08-23 00:00:00	2012-06-07 00:00:00	2011-12-05 00:00:00	2011-11-25 00:00:00	2011-11-25 00:00:00	60.000000	3000.000000	3.047147e+04	9.300000	0.160000
75%	83648.250000	2013-11-07 00:00:00	2014-04-30 00:00:00	2013-12-23 00:00:00	2013-12-23 00:00:00	2014-01-13 00:00:00	90.000000	17039.750000	1.664471e+05	23.592500	0.470000
max	86823.000000	2015-12-01 00:00:00	2015-08-24 00:00:00	2015-12-31 00:00:00	2015-09-14 00:00:00	2015-09-14 00:00:00	1000.000000	619999.000000	5.951990e+06	1345.640000	238.650000
std	31944.332496	NaN	NaN	NaN	NaN	NaN	76.579764	40035.302961	3.452921e+05	45.609223	3.275808

Shipment & Delivery Analysis

Shipment analysis was performed to understand transportation trends, delivery performance, and operational shipment distribution.

Shipment Mode Analysis

Shipment mode distribution analysis showed that:

- AIR transportation dominates shipment operations
- TRUCK transportation is the second most commonly used mode
- AIR CHARTER and OCEAN shipments are comparatively lower

This indicates that the organization prioritizes faster transportation for healthcare products.

Delivery Performance Analysis

Delivery status analysis showed:

- most shipments were delivered on time
- many shipments were delivered earlier than scheduled
- delayed shipments still exist within the supply chain

The average Delay Days value was negative, indicating that many shipments arrived before their scheduled delivery dates.

Country-wise Shipment Analysis

Country shipment analysis revealed that:

- South Africa

- Nigeria
- Côte D'Ivoire

received the highest number of shipments.

These countries represent major operational hubs within the supply chain network.

Monthly Shipment Trend

Monthly shipment analysis showed fluctuations in shipment activity across months and years.

This helps identify:

- operational growth trends
- shipment demand patterns
- transportation planning opportunities



Vendor Analysis

Vendor analysis was performed to evaluate shipment contribution, delivery reliability, and operational dependency on suppliers.

Top Vendors Analysis

Vendor shipment volume analysis showed that:

- SCMS from RDC contributes the highest shipment volume
- shipment operations heavily depend on a few major vendors

This indicates operational concentration within the supply chain network.

Vendor Delay Performance

Vendor delay analysis revealed that:

- some vendors contribute to higher shipment delays
- delivery performance varies significantly across vendors

This analysis helps identify vendors requiring operational monitoring.

Vendor Shipment Mode Distribution

Shipment mode analysis by vendor showed how vendors utilize different transportation methods.

This helps understand:

- transportation strategy
- shipment urgency
- operational behavior

Vendor Freight Contribution

Vendor freight contribution analysis showed that a few vendors contribute significantly to overall freight spending.

This helps evaluate:

- transportation cost concentration
- vendor-related operational expenses



Freight & Product Analysis

Freight and product analysis was performed to evaluate transportation costs, operational efficiency, and product contribution.

Freight Cost by Shipment Mode

Freight cost analysis showed that:

- AIR CHARTER shipments are the most expensive
- OCEAN shipments also have high transportation costs
- TRUCK transportation has comparatively lower freight costs

This helps identify expensive transportation methods.

Freight Cost Distribution

Freight cost distribution analysis showed:

- significant variation in freight costs
- presence of high-cost shipment outliers
- operational cost complexity

Weight vs Freight Cost Analysis

Scatter plot analysis revealed:

- shipment weight has only a weak relationship with freight cost
- other factors such as shipment mode and destination also influence pricing

Product Group Analysis

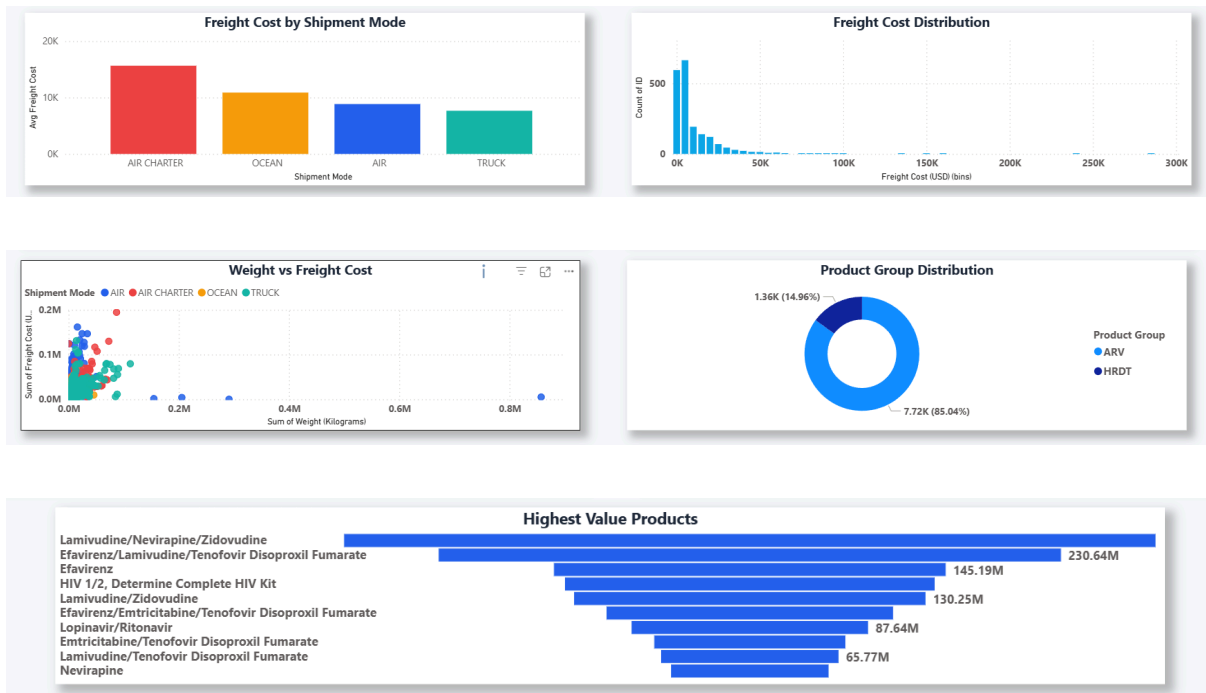
Product analysis revealed that:

- ARV products dominate shipment operations
- HRDT products contribute a smaller shipment share

Highest Value Products

A limited number of healthcare products contribute significantly to shipment value and operational spending.

This helps identify critical products within the healthcare supply chain.



Data Analysis using SQL

SQL was used to perform structured business analysis using MySQL.

The cleaned dataset was imported into a MySQL database for querying and operational analysis.

The following business analyses were performed:

- total shipment analysis
- shipment mode analysis
- delay analysis
- country-wise shipment analysis
- vendor analysis
- freight cost analysis
- product analysis

Database Creation

```
CREATE DATABASE supply_chain_db;
USE supply_chain_db;
```

Example SQL Queries

```
SELECT COUNT(*) FROM shipments;
```

```
SELECT Country, COUNT(*)
```

FROM shipments
GROUP BY Country;

```
SELECT `Shipment Mode`,  
AVG(`Freight Cost (USD)`)  
FROM shipments  
GROUP BY `Shipment Mode`;
```

SQL analysis helped generate operational business insights and KPI calculations.

```
1 • CREATE DATABASE supply_chain_db;  
2 • USE supply_chain_db;  
3 • SELECT * FROM shipments LIMIT 30;  
.
```

ID	Project Code	PQ #	PO / SO #	ASN/DN #	Country	Managed By	Fulfill Via	Vendor INCO Term	Shipment Mode	PQ First Sent to Client Date	PO Sent to Vendor Date	Scheduled Delivery Date	Deliv Date
1	100-CI-T01	Pre-PQ Process	SQMS-4	ASN-8	Côte d'Ivoire	PMO - US	Direct Drop	EXW	AIR			2006-06-02	2006-
3	108-VN-T01	Pre-PQ Process	SQMS-13	ASN-85	Vietnam	PMO - US	Direct Drop	EXW	AIR			2006-11-14	2006-
4	100-CI-T01	Pre-PQ Process	SQMS-20	ASN-14	Côte d'Ivoire	PMO - US	Direct Drop	FCA	AIR			2006-08-27	2006-
15	108-VN-T01	Pre-PQ Process	SQMS-78	ASN-50	Vietnam	PMO - US	Direct Drop	EXW	AIR			2006-09-01	2006-
16	108-VN-T01	Pre-PQ Process	SQMS-81	ASN-55	Vietnam	PMO - US	Direct Drop	EXW	AIR			2006-08-11	2006-
23	112-NG-T01	Pre-PQ Process	SQMS-87	ASN-57	Nigeria	PMO - US	Direct Drop	EXW	AIR			2006-09-28	2006-
44	110-ZM-T01	Pre-PQ Process	SQMS-139	ASN-130	Zambia	PMO - US	Direct Drop	DDU	AIR			2007-01-08	2007-
45	109-TZ-T01	Pre-PQ Process	SQMS-140	ASN-94	Tanzania	PMO - US	Direct Drop	EXW	AIR			2006-11-24	2006-
46	112-NG-T01	Pre-PQ Process	SQMS-156	ASN-93	Nigeria	PMO - US	Direct Drop	EXW	AIR			2006-12-07	2006-
47	110-ZM-T01	Pre-PQ Process	SQMS-165	ASN-199	Zambia	PMO - US	Direct Drop	CIP	AIR		2006-11-13	2007-01-30	2007-
60	110-ZM-T01	Pre-PQ Process	SQMS-221	ASN-223	Zambia	PMO - US	Direct Drop	CIP	AIR		2006-12-01	2007-02-16	2007-

```
7 • SELECT SUM(`Freight Cost (USD)`) AS total_freight_cost  
8 FROM shipments;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
total_freight_cost			
93035043.5900037			


```

10 • SELECT
11     `Shipment Mode`,
12     AVG(`Freight Cost (USD)`) AS avg_freight_cost
13 FROM shipments
14 GROUP BY `Shipment Mode`
15 ORDER BY avg_freight_cost DESC;
16

```

Result Grid		 Filter Rows:	Export: 	Wrap Cell Content: 
	Shipment Mode	avg_freight_cost		
▶	AIR CHARTER	15773.308476923025		
	OCEAN	11086.598611859814		
	AIR	8812.804518696124		
	TRUCK	7644.096872791633		

```
17 • SELECT
18     Country,
19     COUNT(*) AS shipment_count
20 FROM shipments
21 GROUP BY Country
22 ORDER BY shipment_count DESC
23 LIMIT 10;
24
```

Result Grid



Filter Rows:

Export:



	Country	shipment_count
►	South Africa	1406
	Nigeria	1194
	Côte D'Ivoire	1083
	Uganda	779
	Vietnam	688
	Zambia	683
	Haiti	655
	Mozambique	631
	Zimbabwe	538
	Tanzania	518

```

25 • SELECT
26     Country,
27     AVG(`Delay Days`) AS avg_delay
28 FROM shipments
29 GROUP BY Country
30 ORDER BY avg_delay DESC
31 LIMIT 10;
32

```

Result Grid



Filter Rows:

Export:

	Country	avg_delay
▶	Congo, Drc	11.2402
	Togo	5.3333
	Benin	2.3846
	Senegal	1.0000
	Kenya	0.8919
	Pakistan	0.0000
	Kyrgyzstan	0.0000
	Lesotho	0.0000
	Afghanistan	0.0000
	Angola	0.0000

```

33 • SELECT
34     Vendor,
35     COUNT(*) AS shipment_count
36 FROM shipments
37 GROUP BY Vendor
38 ORDER BY shipment_count DESC
39 LIMIT 10;
40

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content

	Vendor	shipment_count
▶	SCMS from RDC	5404
	Organics, Ltd	754
	S. BUYS WHOLESALER	715
	Aurobindo Pharma Limited	668
	Trinity Biotech, Plc	355
	ABBVIE LOGISTICS (FORMERLY ABBOTT LOGIS...	347
	PHARMACY DIRECT	326
	MYLAN LABORATORIES LTD (FORMERLY MATRI...	317
	HETERO LABS LIMITED	277
	CIPLA LIMITED	175

```

41 • SELECT
42     Vendor,
43     AVG(`Delay Days`) AS avg_delay
44 FROM shipments
45 GROUP BY Vendor
46 ORDER BY avg_delay DESC
47 LIMIT 10;
48

```

Result Grid   Filter Rows: Export:  Wrap Cell Conte

	Vendor	avg_delay
▶	CIPLA LIMITED	4.7771
	Aurobindo Pharma Limited	4.4117
	Abbott GmbH & Co. KG	2.2857
	BIO-RAD LABORATORIES (FRANCE)	2.1071
	EMCURE PHARMACEUTICALS LTD	1.3902
	SHANGHAI KEHUA BIOENGINEERING CO.,LTD. ...	1.0571
	STRIDES ARCOLAB LIMITED	0.9785
	ASPEN PHARMACARE	0.8780
	Organics, Ltd	0.8382
	Orasure Technologies Inc.	0.5000

```

49 • SELECT
50     `Delivery Status`,
51     COUNT(*) AS total
52 FROM shipments
53 GROUP BY `Delivery Status`;
54

```

Result Grid			Filter Rows:	Export
	Delivery Status	total		
▶	On Time	6323		
	Late	1186		
	Early	2814		

```

55 • SELECT
56     `Product Group`,
57     COUNT(*) AS total_products
58 FROM shipments
59 GROUP BY `Product Group`
60 ORDER BY total_products DESC;
61

```

Result Grid			Filter Rows:	Export
	Product Group	total_products		
▶	ARV	8550		
	HRDT	1727		
	ANTM	22		
	ACT	16		
	MRDT	8		

```

62 • SELECT
63     `Molecule/Test Type`,
64     SUM(`Line Item Value`) AS total_value
65 FROM shipments
66 GROUP BY `Molecule/Test Type`
67 ORDER BY total_value DESC
68 LIMIT 10;
69

```

Result Grid   Filter Rows: Export:  Wrap Cell Cor		
	Molecule/Test Type	total_value
▶	Lamivudine/Nevirapine/Zidovudine	300755250.15000004
	Efavirenz/Lamivudine/Tenofovir Disoproxil Fuma...	230639842.14000005
	Efavirenz	145187675.30000013
	HIV 1/2, Determine Complete HIV Kit	137022306.8
	Lamivudine/Zidovudine	130253539.28999996
	Efavirenz/Emtricitabine/Tenofovir Disoproxil Fu...	106190100.38000001
	Lopinavir/Ritonavir	87643675.52999997
	Emtricitabine/Tenofovir Disoproxil Fumarate	70831172.16000001
	Lamivudine/Tenofovir Disoproxil Fumarate	65768346.98000005
	Nevirapine	58366415.639999956

```

70 • SELECT
71     Country,
72     AVG(`Freight Cost (USD)`) AS avg_freigh
73 FROM shipments
74 GROUP BY Country
75 ORDER BY avg_freight DESC
76 LIMIT 10;
77

```

Result Grid



Filter Rows:

Export:



	Country	avg_freight
▶	Cameroon	25384.999133333346
	Rwanda	14797.827046511647
	Nigeria	14064.072060301398
	Kenya	13931.574234234238
	Sierra Leone	12059.46
	Ghana	11150.749310344829
	Zambia	10665.885710102522
	Tanzania	10530.965357142872
	Ethiopia	10452.68534722222
	Zimbabwe	10415.344433085545


```
78 • SELECT
79     `Shipment Mode`,
80     COUNT(*) AS total_shipments
81 FROM shipments
82 GROUP BY `Shipment Mode`;
```

Result Grid  Filter Rows: Export		
	Shipment Mode	total_shipments
▶	AIR	6472
	TRUCK	2830
	AIR CHARTER	650
	OCEAN	371

Dashboard in Power BI

An interactive Power BI dashboard was created to visualize shipment performance, vendor performance, freight analysis, delivery analysis, and product contribution.

The dashboard provides a clean and interactive representation of key operational insights.

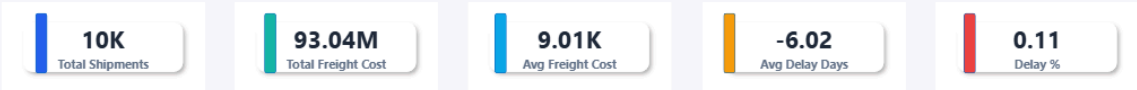
The dashboard was designed using:

- bookmarks
- buttons
- slicers
- interactive visuals
- grouped analysis sections

The dashboard contains three major sections:

1. Shipment & Delivery Analysis
 2. Vendor Analysis
 3. Freight & Product Analysis
-

Global Supply Chain Shipment Performance Dashboard



Shipment & Delivery Analysis

Vendor Analysis

Freight & Product Analysis

Country

All

Shipment Mode

All

Vendor

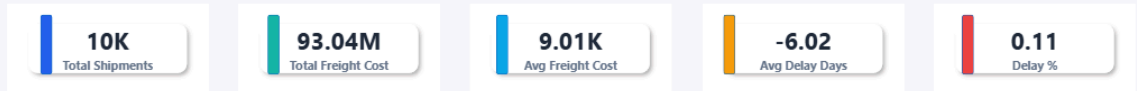
All

Product Group

All



Global Supply Chain Shipment Performance Dashboard



Shipment & Delivery Analysis

Vendor Analysis

Freight & Product Analysis

Country

All

Shipment Mode

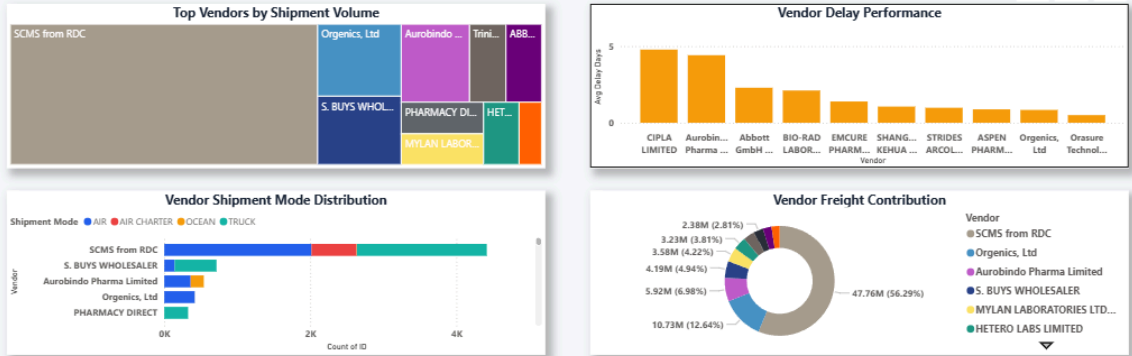
All

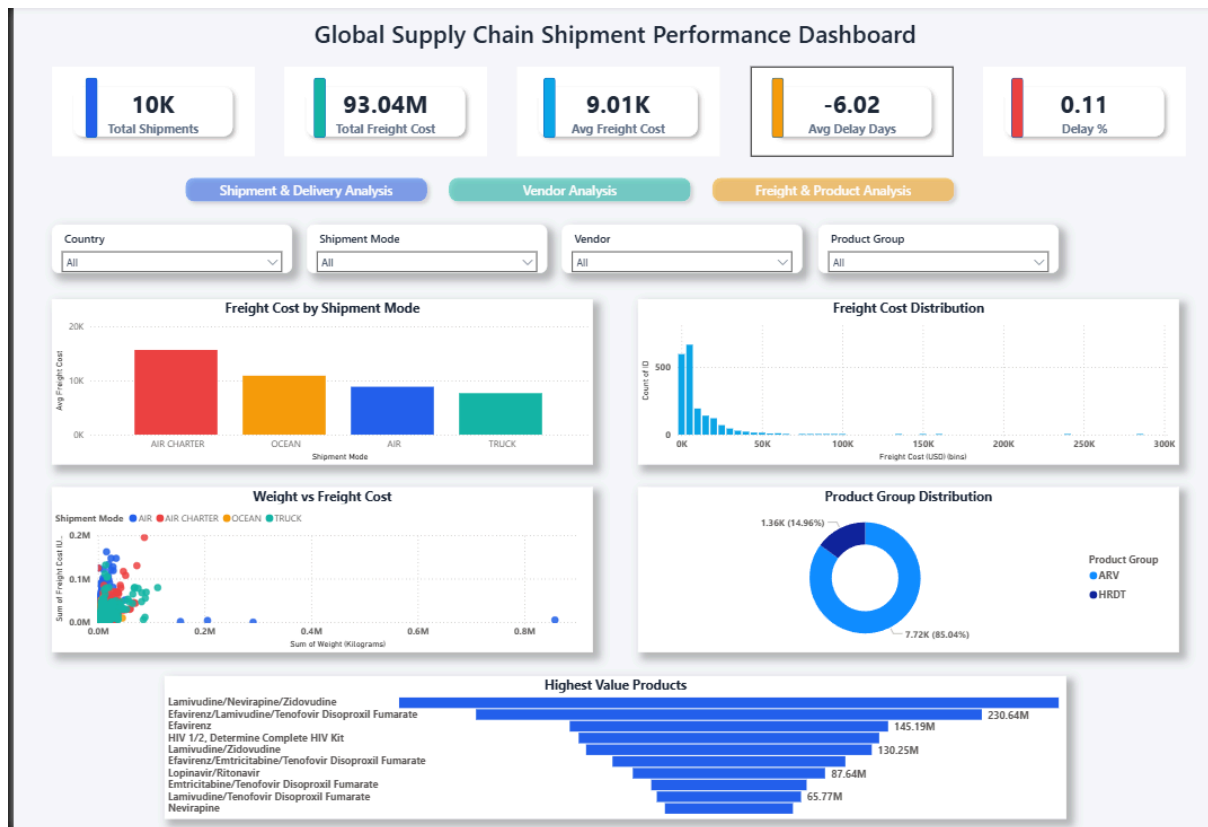
Vendor

All

Product Group

All





Key Features of the Dashboard

KPI Cards

The dashboard includes:

- Total Shipments
- Total Freight Cost
- Average Freight Cost
- Average Delay Days
- Delay Percentage

These KPIs provide a quick overview of operational supply chain performance.

Interactive Navigation

Bookmarks and buttons were used to switch between:

- Shipment Analysis
- Vendor Analysis
- Freight & Product Analysis

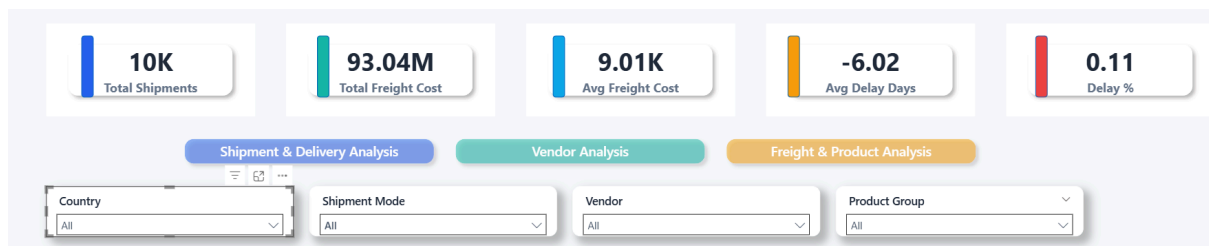
This improved dashboard interaction and reduced visual clutter.

Dynamic Filtering

Slicers were added for:

- Country
- Shipment Mode
- Vendor
- Product Group

These slicers dynamically filter all visuals.



Dashboard Design Approach

The dashboard was designed using a clean and business-focused layout.

Design principles followed:

- one-page dashboard design
- minimal visual clutter
- grouped analytical sections
- consistent color theme
- interactive navigation
- proper spacing and alignment

The design focuses on:

- operational storytelling
 - management reporting
 - business insights
 - user-friendly interaction
-

Insights from Dashboard

1. South Africa, Nigeria, and Côte D'Ivoire are the largest shipment destinations.
 2. AIR transportation dominates shipment operations.
 3. Most shipments are delivered on time, although delayed shipments still exist.
 4. Many shipments are delivered before their scheduled delivery dates.
 5. Certain countries experience significantly higher shipment delays.
 6. Shipment operations depend heavily on a few major vendors.
 7. Some vendors contribute to higher average delivery delays.
 8. Freight costs vary significantly across shipments.
 9. AIR CHARTER shipments are the most expensive transportation method.
 10. Shipment weight has only a weak relationship with freight cost.
 11. ARV products dominate shipment operations.
 12. A limited number of healthcare products contribute significantly to shipment value.
-

Business Recommendations

1. Improve Vendor Monitoring

Monitor vendors with higher shipment delays and improve operational coordination.

2. Optimize Freight Costs

Reduce excessive use of expensive shipment methods such as AIR CHARTER.

3. Strengthen Delay Management

Improve transportation planning for delay-prone countries.

4. Improve Shipment Planning

Use shipment trend analysis for better forecasting and resource planning.

5. Diversify Vendor Dependency

Reduce operational dependency on a small number of vendors.

6. Focus on High-Value Products

Prioritize transportation efficiency for high-value healthcare products.

7. Improve Logistics Efficiency

Analyze shipment routes and transportation methods to reduce freight expenses.

Conclusion

This project successfully analyzed global healthcare supply chain shipment data to understand shipment performance, freight cost behavior, transportation efficiency, vendor reliability, and product contribution.

Python was used for:

- data cleaning
- preprocessing
- feature engineering
- exploratory data analysis

SQL was used for:

- structured querying
- operational analysis
- KPI generation
- business insights

Power BI was used to create an interactive dashboard for visualizing operational supply chain performance.

The analysis revealed important insights related to:

- shipment delays
- freight cost variability
- transportation methods
- vendor dependency
- delivery performance
- product contribution

The dashboard provides a comprehensive view of the healthcare supply chain system and supports data-driven operational decision-making.

Overall, this project demonstrates how data analytics techniques can be used to solve real-world logistics and supply chain challenges.
